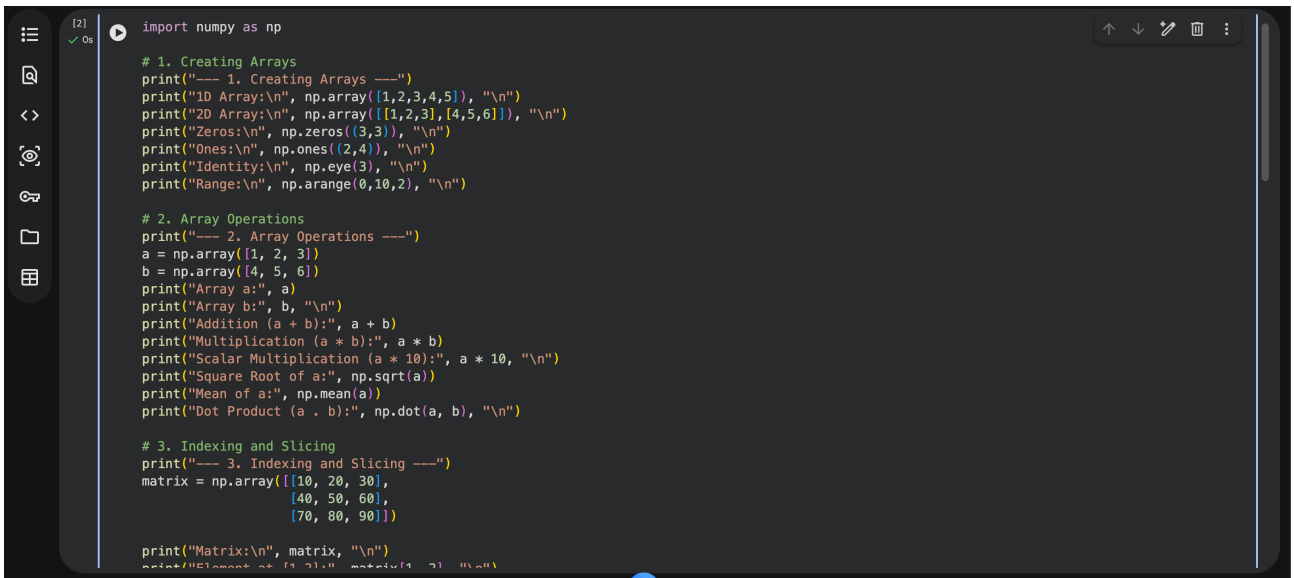
Experiment 1: Creating DataFrames, Selection and Filtering Using Pandas

Aim:

To create a Pandas DataFrame and perform data exploration, column selection, row filtering, and indexing using `head()`, `info()`, `describe()`, `iloc[]`, and `loc[]`.

Algorithm:

1. Import the Pandas library using `import pandas as pd`.
2. Create a dictionary containing Name, Age, and City data.
3. Convert the dictionary into a DataFrame using `pd.DataFrame()`.
4. Display the first few rows using `df.head()`.
5. Display the DataFrame information using `df.info()` and statistical summary using `df.describe()`.
6. Select the Age column using `df['Age']`.
7. Filter rows where Age is greater than 25.
8. Access the first row using `df.iloc[0]`.
9. Access a specific value using `df.loc[0, 'Name']`.
10. Display all the results



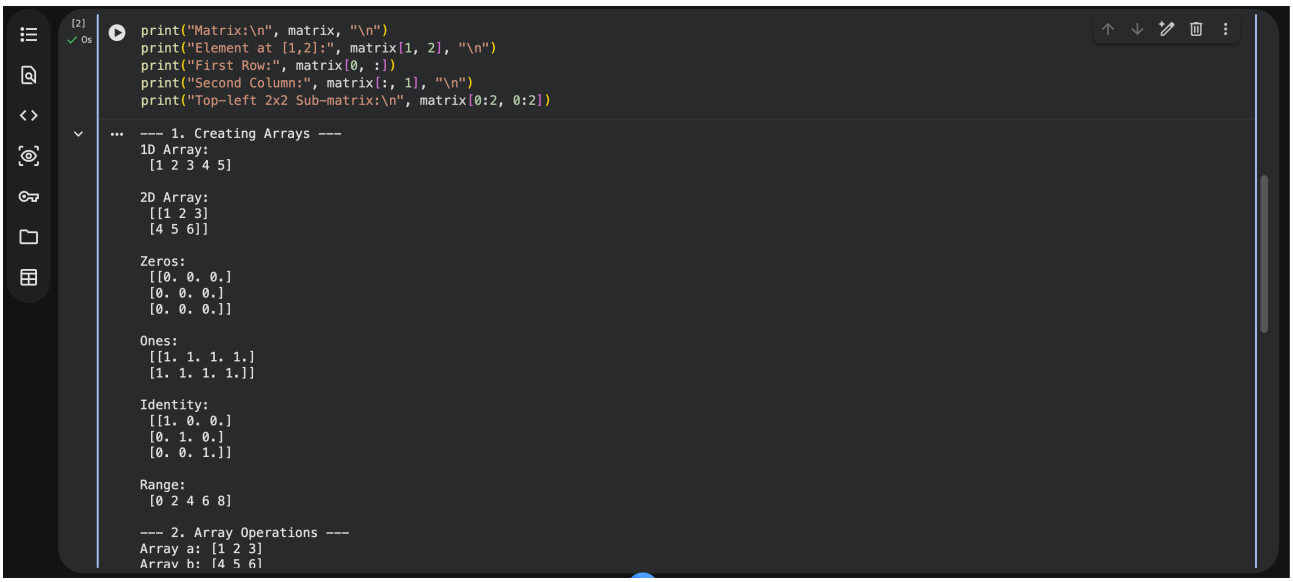
```
import numpy as np

# 1. Creating Arrays
print("--- 1. Creating Arrays ---")
print("1D Array:\n", np.array([1,2,3,4,5]), "\n")
print("2D Array:\n", np.array([[1,2,3],[4,5,6]]), "\n")
print("Zeros:\n", np.zeros((3,3)), "\n")
print("Ones:\n", np.ones((2,4)), "\n")
print("Identity:\n", np.eye(3), "\n")
print("Range:\n", np.arange(0,10,2), "\n")

# 2. Array Operations
print("--- 2. Array Operations ---")
a = np.array([1, 2, 3])
b = np.array([4, 5, 6])
print("Array a:", a)
print("Array b:", b, "\n")
print("Addition (a + b):", a + b)
print("Multiplication (a * b):", a * b)
print("Scalar Multiplication (a * 10):", a * 10, "\n")
print("Square Root of a:", np.sqrt(a))
print("Mean of a:", np.mean(a))
print("Dot Product (a . b):", np.dot(a, b), "\n")

# 3. Indexing and Slicing
print("--- 3. Indexing and Slicing ---")
matrix = np.array([[10, 20, 30],
                   [40, 50, 60],
                   [70, 80, 90]])

print("Matrix:\n", matrix, "\n")
```



```
[2]
0s
print("Matrix:\n", matrix, "\n")
print("Element at [1,2]:", matrix[1, 2], "\n")
print("First Row:", matrix[0, :])
print("Second Column:", matrix[:, 1], "\n")
print("Top-left 2x2 Sub-matrix:\n", matrix[0:2, 0:2])

--- 1. Creating Arrays ---
1D Array:
 [1 2 3 4 5]

2D Array:
 [[1 2 3]
 [4 5 6]]

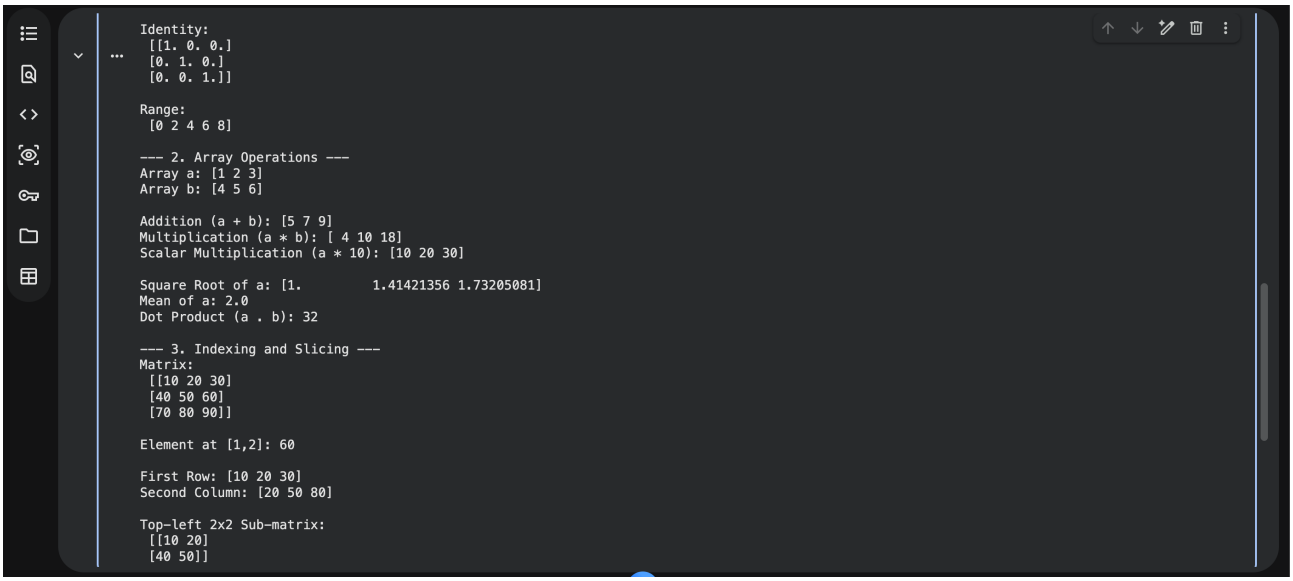
Zeros:
 [[0. 0. 0.]
 [0. 0. 0.]
 [0. 0. 0.]]

Ones:
 [[1. 1. 1.]
 [1. 1. 1.]]

Identity:
 [[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]

Range:
 [0 2 4 6 8]

--- 2. Array Operations ---
Array a: [1 2 3]
Array b: [4 5 6]
```



```
Identity:
 [[1. 0. 0.]
 [0. 1. 0.]
 [0. 0. 1.]]

Range:
 [0 2 4 6 8]

--- 2. Array Operations ---
Array a: [1 2 3]
Array b: [4 5 6]

Addition (a + b): [5 7 9]
Multiplication (a * b): [ 4 10 18]
Scalar Multiplication (a * 10): [10 20 30]

Square Root of a: [1.         1.41421356 1.73205081]
Mean of a: 2.0
Dot Product (a . b): 32

--- 3. Indexing and Slicing ---
Matrix:
 [[10 20 30]
 [40 50 60]
 [70 80 90]]

Element at [1,2]: 60

First Row: [10 20 30]
Second Column: [20 50 80]

Top-left 2x2 Sub-matrix:
 [[10 20]
 [40 50]]
```

Result:

The Pandas DataFrame was created successfully, and data exploration, statistical analysis, column selection, row filtering, and indexing operations were performed successfully.